Problem Chosen: **A** **2026 HiMCM Summary Sheet** Team Control Number: **TEAM_NUMBER**

Floors First, Then Freedom: Ranking 25 U.S. Cities for High Schoolers Who Cannot Rely on a Car

What makes a city great for a high schooler? We define the best city as the one that, for a family on a fixed budget, clears basic floors on rent, violent crime, traffic deaths and air quality, and then gives a public-school student who cannot depend on car rides the most real access to education, health support, independent travel, activities and fair treatment. The city must also keep that lead when reasonable people weigh these things differently.

Criteria and data. We built 20 measures in seven separate domains: education, safety, health, mobility, activities, affordability and inclusion. Affordability and inclusion are deliberately kept apart. Every measure is computed from public data for the 25 most populous U.S. cities (Census Vintage 2025): CRDC and EDFacts school records, FBI and NHTSA FARS, EPA air data, SAMHSA, EPA Smart Location Database, FTA NTD, HUD, BEA and HRC. We dropped nine candidate criteria (for example weather and college rankings) and explain why (Section 3.2).

Model. (1) Four non-compensable floors screen cities before any scoring. (2) Each measure is converted to value by a piecewise-linear curve with fixed, city-independent reference points. (3) Linear programs test every pair of cities over the whole set of allowed preference weights. (4) Stochastic multicriteria acceptability analysis (SMAA) repeats the ranking 10,000 times under random weights and data errors. (5) Minimax regret and ELECTRE III check a cautious decision rule and a no-weak-spot rule.

Results. Seattle is our Best City for a High Schooler.

- Under the preferences stated in our definition, Seattle ranks first in **63.6%** of 10,000 simulations and in the top three in **97.9%**.
- It has the smallest worst-case regret (**0.154** on a 0–1 value scale) and passes all four floors.
- Denver is second (first in 23.5% of simulations) and narrowly leads on equal domain weights (0.8210 against 0.8205). Chicago is the most stable city when no weak spot may be traded away.
- Among cities that pass the floors, the top five are Seattle, Denver, New York, Chicago and Austin. San Francisco would place fifth but fails the rent floor for an $80,000 family (rent takes 50.4% of income).

Significant revisions. Auditing real data changed our model twice.

- M0 → M1: a transit-frequency measure was blank for 96% of San Diego's residents, and one adult charter school pulled San Francisco's graduation rate to 56%. Fixing both moved San Francisco from 13th to 8th.
- M1 → M2: we replaced transit service hours with jobs reachable within 45 minutes by scheduled transit, so we measure what a student can actually reach.

Why it is more than our opinion. Reference points are fixed, so removing any city reverses **0** pairs of other cities (an entropy-weight ranking reverses 90). Seattle, Denver and Chicago remain the leaders under equal weights, data noise up to four times its size, ±10% node shifts and five alternative aggregation rules; under our stated preferences Seattle stays first with node shifts (60%) and without the two statements that limit affordability (56%). School-age survey outcomes agree with our domain scores: skipping school for fear of being unsafe rises with violent crime ($\rho = +0.73$), and sadness or hopelessness correlates with our health domain ($\rho = -0.76$).

Caveat. Seattle's lead belongs to families who can afford its neighborhoods. Only 38.5% of its residents live in census tracts that are affordable on $80,000 and well served by transit. For a $50,000 family the best choices are Chicago and Columbus.

Keywords: piecewise-linear value model; robust ordinal regression; linear programming; SMAA; minimax regret; ELECTRE III; transit accessibility

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1. Introduction

1.1 Background

More than 15 million students attend U.S. public high schools. Most city rankings are written for adults: they weigh job markets, nightlife and home prices. A high schooler's life runs on different things: whether the nearest public school offers advanced courses, whether a bus can get her to practice and back after school, whether a counselor or a clinic will see a 16-year-old, and whether she is treated fairly. A teenager usually cannot drive and cannot choose the family budget, so the city's value to her depends on what she can reach and afford without an adult's car.

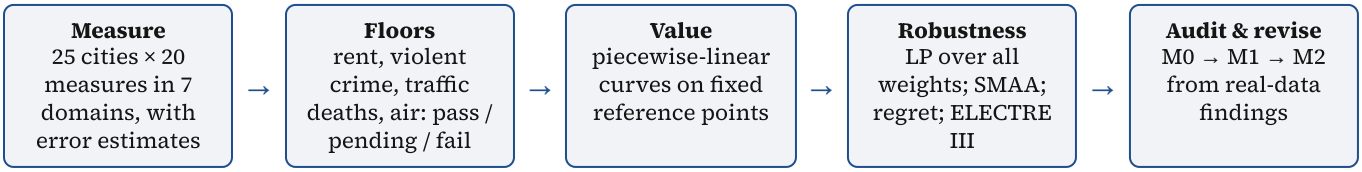
1.2 The task and our reading of it

1. Choose and justify criteria for a "best city for a high schooler", and discuss criteria left out.
2. Gather and cite data, apply the model, and discuss limitations.
3. Rank at least ten cities and reach a reasoned conclusion about the best one, under our own definition.
4. Show why the ranking has merit beyond opinion, and document at least one significant revision.
5. Write a one-to-two-page non-technical news article that includes our modeling process.

Our definition of "best". *The best city is the one that, for a family on a fixed budget, clears basic floors on cost and serious risk, and then gives a public-school student who cannot depend on car rides the most real access to education, health support, independent travel, activities and fair treatment — and keeps that lead across reasonable disagreements about what matters most.*

Three features of this definition shape the model. **Floors** come before trade-offs: a low rent should not buy back a dangerous street. **Real access** means measuring what a student can use (enrollments in advanced courses, jobs reachable by transit in 45 minutes), not how many buildings exist. **Robustness** means the answer must hold over a whole set of reasonable weightings, not a single set we chose.

1.3 Our approach



The candidate pool is fixed before any score is computed: the 25 most populous U.S. cities in the Census Vintage 2025 estimates [1]. Every number in this paper is reproduced by our code from archived source files.

2. Assumptions

1. **The student attends a public high school and cannot rely on daily car rides.** *Justification:* most U.S. high schoolers attend public schools, and many teenagers do not drive; the problem asks about high schoolers, not families with cars.
2. **City-wide statistics describe the conditions a typical resident student meets.** *Justification:* consistent neighborhood data are not available for all measures in 25 cities. We test this assumption with census-tract data in Section 9.3.

3. **The reference family earns $80,000 a year and rents a two-bedroom home.** *Justification:* this is close to the U.S. median household income; we scan $50,000–$120,000 in Section 9.2.
4. **Recent multi-year averages represent current conditions.** *Justification:* we use 2021–2025 data and average several years where single years are noisy (crime 2022–2025; traffic and air 2021–2024).
5. **Once floors are met, a better result in one measure can partly make up for a worse one in another.** *Justification:* this is the usual way families trade off. We test it against a non-compensatory method (ELECTRE III) and against low-substitution aggregation (Section 8.2).
6. **Missing values are not zero.** *Justification:* a missing report is not evidence of no service. Two of 500 cells are missing; we show each city's rank with the missing value set to the worst and to the best case.

3. Criteria: What We Measure and What We Leave Out

3.1 Seven domains and twenty measures

We began from the needs of a 14–18-year-old rather than from available data. Each domain answers one question, and the domains stay separate even when their measures are correlated. In particular, affordability (can the family pay?) and inclusion (is the student treated fairly?) are different questions and are never merged. Each measure has four fixed reference points ("nodes"): the worst node is worth 0 and the best node earns the full value of that measure. Node values come from standards where they exist (for example the 250-students-per-counselor recommendation [16] and HUD's 30% and 50% rent-burden lines [12]) and otherwise from round values that span realistic U.S. conditions. They were fixed before any city was scored.

Domain (question)	Measure	Better	Nodes (worst → best)	Source
Education — what can a student learn?	AP, IB and dual-enrollment course enrollments per 100 students	↑	10 / 25 / 40 / 55	CRDC 2023–24 [3]
	Four-year graduation rate, non-charter schools (%)	↑	60 / 75 / 85 / 95	EDFacts 2020–21 [4]
	School counselors per 1,000 students	↑	2 / 3 / 4 / 6	CRDC 2023–24 [3]
Safety — how likely is serious harm?	Violent crimes per 100,000 residents (2022–25)	↓	1200 / 900 / 600 / 300	FBI CDE [5]
	Traffic deaths per 100,000 residents (2021–24)	↓	20 / 14 / 8 / 2	NHTSA FARS [6]
Health — can a student stay well?	Days with AQI above 100 (% of days, 2021–24)	↓	30 / 15 / 5 / 0	EPA AirData [7]
	School nurses, psychologists, social workers per 1,000 students	↑	0.5 / 2 / 4 / 8	CRDC 2023–24 [3]
	Youth mental-health facilities per 100,000 residents under 18	↑	2 / 6 / 12 / 20	SAMHSA 2025 [8]
Mobility — can a student get around alone?	Jobs reachable in 45 min by transit and walking (thousands)	↑	10 / 60 / 200 / 600	EPA SLD v3 [9]
	National Walkability Index (1–20)	↑	5.75 / 10.5 / 15.25 / 20	EPA SLD v3 [9]
	Student fare for one local ride (USD)	↓	2 / 1 / 0.5 / 0	operator fare pages [10]
	Weekend service as a share of weekday service	↑	0.3 / 0.5 / 0.7 / 0.9	FTA NTD 2022–25 [11]
Activities — what can a student do after school?	Residents within a 10-minute walk of a park (%)	↑	30 / 55 / 80 / 100	TPL ParkScore 2026 [13]
	Teen library program attendance per 1,000 residents	↑	0 / 10 / 25 / 50	IMLS FY2023–24 [14]
	School sports participations per 100 students	↑	10 / 20 / 35 / 50	CRDC 2023–24 [3]
	Minimum wage for ages 16–17, adjusted for local prices (USD/h)	↑	6.5 / 10 / 14 / 18	DOL, state laws [15]; BEA [2]
Affordability — can the family pay?	Two-bedroom Fair Market Rent as % of $80,000 income	↓	50 / 40 / 30 / 15	HUD FY2024 [12]
	Prices of goods and other services (U.S. = 100)	↓	110 / 105 / 100 / 95	BEA RPP 2024 [2]
Inclusion — is the student treated fairly?	City non-discrimination law and services (% of HRC points)	↑	25 / 50 / 75 / 100	HRC MEI 2025 [17]
	Students suspended out of school per 100	↓	16 / 10 / 5 / 0	CRDC 2023–24 [3]

Avoiding double counting. Counselors appear only in education; nurses, psychologists and social workers only in health. Housing is excluded from the BEA price measure because rent is measured directly. Transit service and transit access are not both scored: in M2 the access measure already contains the service it uses (Section 7).

3.2 Criteria we considered and left out

Criterion	Reason left out
Weather and climate	A matter of taste, not of opportunity; it belongs in a personal profile, not in the definition of "best"
College and university rankings	Measures adult institutions; what reaches a student is captured by dual-enrollment courses
Test scores (grades 3–8, SEDA)	Wrong age range; license does not allow redistribution
Nightlife, restaurants, job growth	Weakly related to ages 14–18
Teen employment rate	Mixes opportunity with need: a high rate can mean students must work
Racial and ethnic diversity	Describes who lives there, not how students are treated; kept as background only
Walk Score, AllTransit	Paid licenses; replaced by open EPA measures
Youth survey outcomes (YRBS)	Available for only 12 of 25 cities; used to validate the model instead (Section 8.4)
Adult health outcomes (CDC PLACES)	Measures adults; teenagers' needs differ

3.3 The candidate cities

The pool is the 25 most populous cities in the Census Vintage 2025 estimates [1], using the Census definition of each city (for Indianapolis and Nashville, the consolidated "balance"). The list was frozen before any data were scored, so no city was added or dropped to favor a result.

4. Data

4.1 Sources and processing

We assembled a long table of 157,256 records (121,160 numeric values) from 28 public sources. It covers multiple years, geographic levels (school, police agency, county, metro area, transit operator) and sources [1]–[17]. From it we computed the 25 × 20 matrix used by the model. The main processing rules:

- **School measures** sum all high-school-offering schools located inside each city boundary (2,264 schools in CRDC 2023–24) and divide by their enrollment. Negative CRDC reserve codes are treated as missing, never as zero.
- **Rates** use Census city population, except where the police agency covers a larger area (Charlotte-Mecklenburg, Las Vegas Metropolitan and Metro Nashville), where we use the FBI agency population.
- **Area measures** (air quality, rent) are weighted by the share of the city's residents in each county.
- **Neighborhood measures** (walkability, transit access) are population-weighted means over the census block groups inside each city (22,398 populated block groups).

4.2 Measurement error

Every value carries a standard deviation used in the uncertainty analysis:

- **Multi-year measures** (crime, traffic, air): the standard error of the annual mean.
- **School measures**: the difference between the two CRDC waves (2021–22 and 2023–24) divided by $\sqrt{2}$, the sample standard deviation of two values.
- **Graduation rates** published as ranges: the range width converted to a 90% interval.
- **Counts** (youth mental-health facilities): the Poisson standard deviation $\sqrt{n}$.

Policy values (fares, wages, HRC points) are treated as exact.

4.3 Missing data

Two of the 500 cells are missing:

- San Francisco reports no school athletics in schools enrolling 94% of its students, which we treat as a reporting gap.
- Washington, D.C. is not rated by HRC.

The reference ranking fills a missing cell with the 25-city median and reports the rank range from worst to best fill. San Francisco's rank moves between 6th and 10th; Washington's between 9th and 17th. All other ranks are unaffected.

5. The Model

5.1 Notation

Symbol	Meaning
i, k	cities, $i = 1, \dots, 25$
j	measures, $j = 1, \dots, 20$; $d(j)$ is the domain of measure j
x_{ij}, s_{ij}	value of measure j in city i and its standard deviation
$t_{j0}, \dots, t_{j3}$	fixed nodes of measure j (worst to best)
$\phi_{ijl} \in [0, 1]$	how much of segment l of measure j city i covers
$\theta_{jl} \geq 0$	value of segment l of measure j; θ collects all 60 of them
Θ	set of allowed preference vectors; θ^0 is the reference vector
$U_i(\theta)$	overall value of city i, between 0 and 1

5.2 Step 1: floors that cannot be bought back

Some conditions are needs, not preferences. A city fails if any of four raw values crosses a floor:

$$F = \{\, i : \ \text{rent}_i \leq 50\%\ \text{of income},\ \ \text{violent crime}_i \leq 1200,\ \ \text{traffic deaths}_i \leq 20,\ \ \text{bad-air days}_i \leq 30\% \,\}.$$

The rent floor is HUD's "severe burden" line [12]. The other three floors equal the worst node of each measure; no city's multi-year average crosses them, and three cities come within measurement error. A floor is checked on the raw value, before any weighting, so no other strength can offset it.

Because each value has an error, a city has three states for each floor. It **passes** if its 90% interval $x \pm 1.645s$ lies fully on the safe side, **fails** if it lies fully on the wrong side, and is **pending** otherwise. The joint chance of passing all four floors is estimated by 20,000 simulated data sets. Floors were also scanned over ranges (Section 9.2), because no floor value is universal.

5.3 Step 2: a piecewise-linear value for each measure

Families value the first improvements most. Going from 10% to 25% of students in advanced courses matters more than going from 40% to 55%. We model this with a curve made of three straight segments between the four fixed nodes. The coverage of segment l is

$$\phi_{ijl} = \min\left(1,\ \max\left(0,\ \frac{x_{ij} - t_{jl}}{t_{j,l+1} - t_{jl}}\right)\right), \qquad l = 0, 1, 2,$$

(with the inequality reversed for "lower is better" measures), and the overall value is a sum:

$$U_i(\theta) = \sum_{j=1}^{20} \sum_{l=0}^{2} \theta_{jl}\, \phi_{ijl} = \phi_i^{\top} \theta.$$

Here θ_{jl} is the value earned by completing segment l. A city at or below the worst node earns 0 on that measure; a city at or above the best node earns $\theta_{j0} + \theta_{j1} + \theta_{j2}$. No separate "weight" multiplies the curve, because the segment values already include how much the measure matters. This keeps the score linear in θ, which is what makes exact robustness analysis possible.

Fixed reference points. Nodes are absolute values, not the best and worst city in the pool. Adding or removing a city therefore never changes the order of two other cities.

5.4 Step 3: the set of reasonable preferences

People disagree about how much safety is worth compared with, say, activities. Instead of one weight vector, we keep the whole set Θ of vectors that satisfy:

- **Normalization:** $\theta \geq 0$ and $\sum \theta = 1$, so $U_i \in [0, 1]$.
- **Diminishing returns:** $\theta_{j0} \geq \theta_{j1} \geq \theta_{j2}$ for every measure, and the last segment keeps at least 10% of the measure's value.
- **Balanced domains:** each domain receives between half and double an equal share, $\frac{1}{14} \leq \sum_{j \in D} \theta_j \leq \frac{2}{7}$.
- **No measure ignored:** within a domain, each measure receives at least one third of an equal share of that domain.

The **reference vector** θ^0 gives each domain 1/7 and splits it equally among its measures. Within each measure, it splits the value 0.563 / 0.289 / 0.148 across the three segments. These are the values of the smooth curve $u(z) = (1 - e^{-2z})/(1 - e^{-2})$ at $z = \frac{1}{3}, \frac{2}{3}, 1$, a moderate diminishing return.

Our stated preferences Θ_S. Our definition in Section 1.2 implies five further statements, each tied to a sentence of the definition:

	Statement	Reason in our definition
P1	Education weight ≥ Affordability weight	The budget is handled by the rent floor; extra savings above it are a softer preference
P2	Mobility weight ≥ Affordability weight	Same reason, for independent travel
P3	Mobility weight ≥ 1/7	The student cannot rely on car rides
P4	Education weight ≥ 1/7	Access to school and courses is the first purpose
P5	Each measure gets 0.6–1.4 times an equal share inside its domain	Every core measure covers its own part of the domain

θ^0 satisfies all five.

5.5 Step 4: what holds for every reasonable preference (linear programs)

For two cities a and b, we solve

$$\underline{\Delta}_{ab} = \min_{\theta \in \Theta} (\phi_a - \phi_b)^\top \theta, \qquad \overline{\Delta}_{ab} = \max_{\theta \in \Theta} (\phi_a - \phi_b)^\top \theta.$$

If $\underline{\Delta}_{ab} \geq 0$, city a is at least as good as b under **every** reasonable preference: a *necessary* preference. If $\overline{\Delta}_{ab} \geq 0$, there is **some** preference under which it is: a *possible* preference. Both are linear programs because Θ is defined by linear inequalities. This is the idea of robust ordinal regression [18].

A city can win only if a **single** vector makes it at least as good as all others at once. We test this by maximizing a margin ε subject to $(\phi_i - \phi_k)^\top \theta \geq \varepsilon$ for all k and $\theta \in \Theta$. City i is a *possible winner* if the best $\varepsilon \geq 0$. Beating each rival with a different weighting is not enough.

Data uncertainty in the LP. Each value of ϕ rises with the measure's quality and $\theta \geq 0$. So, for any θ, the worst case of $U_a - U_b$ over all data inside the 90% intervals puts a at the bad end and b at the

good end. One LP with these two fixed rows therefore gives an exact robust answer for both preference and data uncertainty.

5.6 Step 5: how often each city wins (SMAA)

The LP says what is possible; SMAA [19] says how common it is. We draw 10,000 preference vectors from Θ (or Θ_S) in three stages, each uniform over its allowed range: the seven domain weights, then each domain's split among its measures, then each measure's three segment values. Each draw is paired with a data set in which every value is perturbed by $N(0, s_{ij}^2)$; a missing value is drawn from the observed cities. For each draw we rank all cities. We report

- the share of draws in which each city ranks first, in the top three, and its mean rank;
- the central weight vector of each city's winning draws;
- its *confidence factor*: how often it wins when that central vector is fixed and only the data vary.

With 10,000 independent draws, the standard error of any share is at most 0.005.

5.7 Step 6: two cross-checks with different philosophies

Minimax regret [25] answers a cautious family's question: "If I pick city i and my true preferences are the worst case for that choice, how much value do I lose against the best city?"

$$R_i = \max_{\theta \in \Theta} \left[\max_k U_k(\theta) - U_i(\theta) \right] = \max_k \ \overline{\Delta}_{ki}.$$

It reuses the pairwise LPs; the city with the smallest R_i is the cautious choice.

ELECTRE III [20] does not allow free trade-offs. City a "outranks" b if enough weight of measures supports it (concordance) and no measure shows a far behind (veto). We use indifference, preference and veto thresholds of 0.05, 0.20 and 0.80 on each measure's 0–1 value scale. A veto means a gap of at least 80% of the scale. The standard descending and ascending distillations give two orders; their intersection identifies the *kernel* (cities no other city beats). We also repeat ELECTRE with 400 weight vectors sampled from Θ_S.

5.8 Worked example: Seattle against Denver

Floors. Seattle's two-bedroom rent ($2,645) is 39.7% of $80,000. Violent crime is 777 per 100,000, traffic deaths 5.4 and bad-air days 1.8%. All four 90% intervals lie on the safe side, so Seattle passes with joint probability 1.000.

One measure in detail. Seattle has 54.2 advanced-course enrollments per 100 students. With nodes 10 / 25 / 40 / 55, it covers segments 1 and 2 fully and 14.2/15 = 0.946 of segment 3. With θ^0 (education 1/7 split over three measures, so 0.0476 per measure, divided 0.563 / 0.289 / 0.148), its contribution is

$$0.0476 \times (0.563 + 0.289 + 0.946 \times 0.148) = 0.0472.$$

Denver's 38.6 enrollments cover segment 2 only to 0.906 and earn 0.0393.

All domains. Summing the 20 contributions:

Domain	Education	Safety	Health	Mobility	Activities	Affordability	Inclusion	Total U
Seattle	0.116	0.114	0.119	0.128	0.128	0.078	0.137	**0.8205**
Denver	0.098	0.085	0.131	0.127	0.132	0.115	0.134	**0.8210**
Difference	+0.018	+0.030	−0.012	+0.001	−0.004	−0.037	+0.004	−0.0005

Seattle is safer and stronger in school; Denver is cheaper and has more school health staff and teen library programs.

Robust comparison. The LPs give $U_{\text{Seattle}} - U_{\text{Denver}} \in [-0.301, 0.290]$ over all of Θ and $[-0.151, 0.210]$ over Θ_S. Both contain 0, so the order depends on preferences.

Frequency and caution. Under Θ_S, SMAA puts Seattle first in 63.6% of draws against 23.5% for Denver. Seattle's worst-case regret is 0.154, against 0.210 for Denver.

6. Results: The Ranking

6.1 The reference scorecard

Figure 1 shows each city's value under the reference weights θ^0, split by domain. The top four are separated by 0.032, less than the gap between 4th and 10th place (0.050). Denver and Seattle differ by 0.0005, far below the spread caused by data error alone: under data noise only, each wins about half the time (Denver 52.5%, Seattle 44.5%).

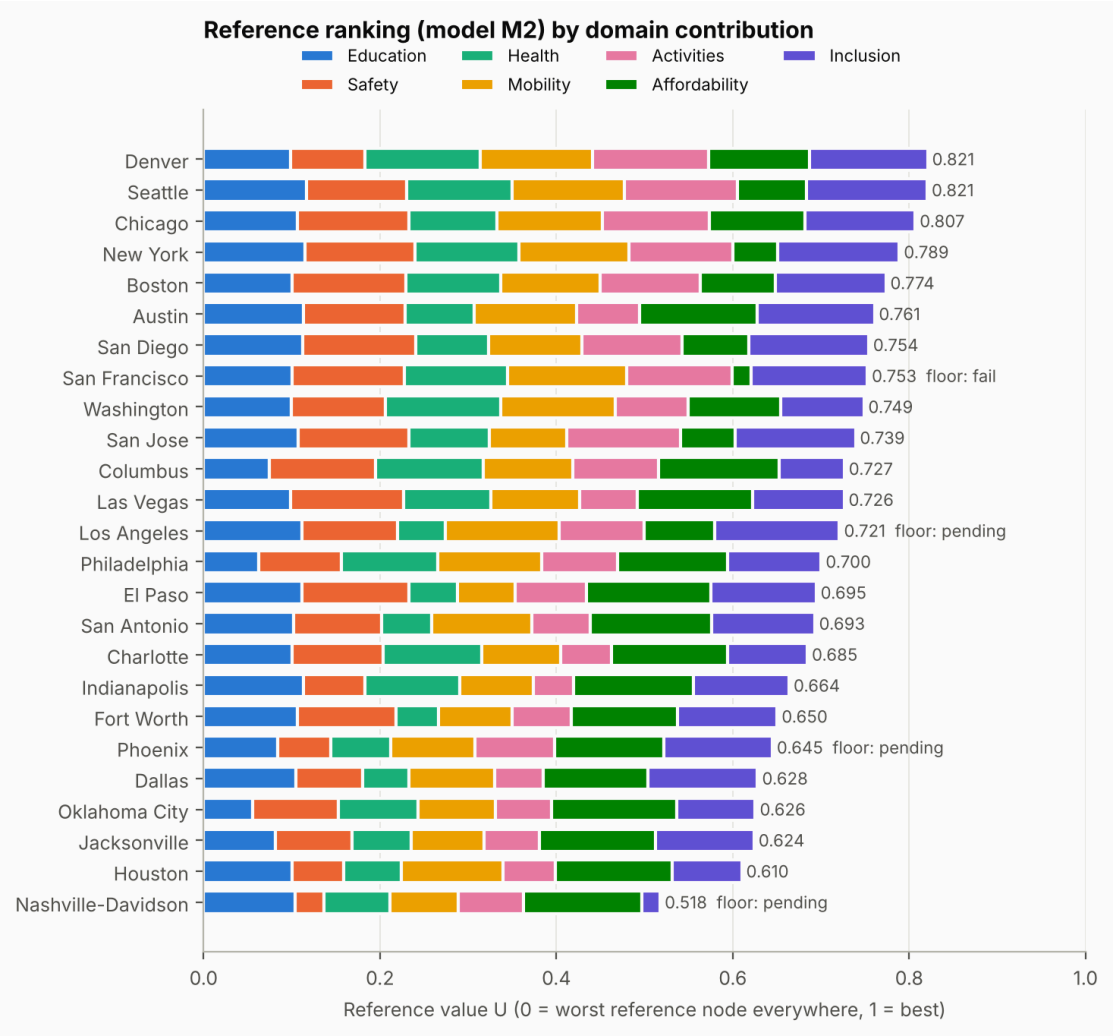


Figure 1. Reference value of each city under equal domain weights, split by domain. Floor status is shown next to the score.

6.2 Floors

At an income of $80,000:

- **21 cities pass all four floors.**
- **San Francisco fails** the rent floor: two-bedroom rent takes 50.4% of income. HUD Fair Market Rents are published standards, so we treat them as exact; San Francisco passes at incomes above

$80,616.

- **Three cities are pending.** In Los Angeles and Phoenix, the share of bad-air days (27.9% and 27.4%) is within error of the 30% floor (joint pass probability 0.81 and 0.60). In Nashville, violent crime of 1,160 per 100,000 is close to 1,200 (0.72).

Pending cities stay in the ranking but are not recommended until confirmed. The joint pass probability from simulation and the conservative Bonferroni bound differ by at most 0.0006, so the bound is not overly cautious here.

6.3 How stable is the ranking? (all reasonable preferences)

Over the full set Θ, the LPs fix the order of **no** pair of cities, and all 25 cities are possible winners. The broad set allows extreme views (one domain at double weight, another at half), and almost any city has a domain in which it excels. Frequency, not possibility, then separates the cities (Figure 2):

- **First place:** Denver 36.9%, Seattle 36.3%, Chicago 13.0%, New York 5.6% and Austin 3.9%.
- **Top three:** Seattle 82.3% and Denver 74.8%.
- **90% rank intervals:** Seattle 1–5, Denver 1–7 and Chicago 1–7.

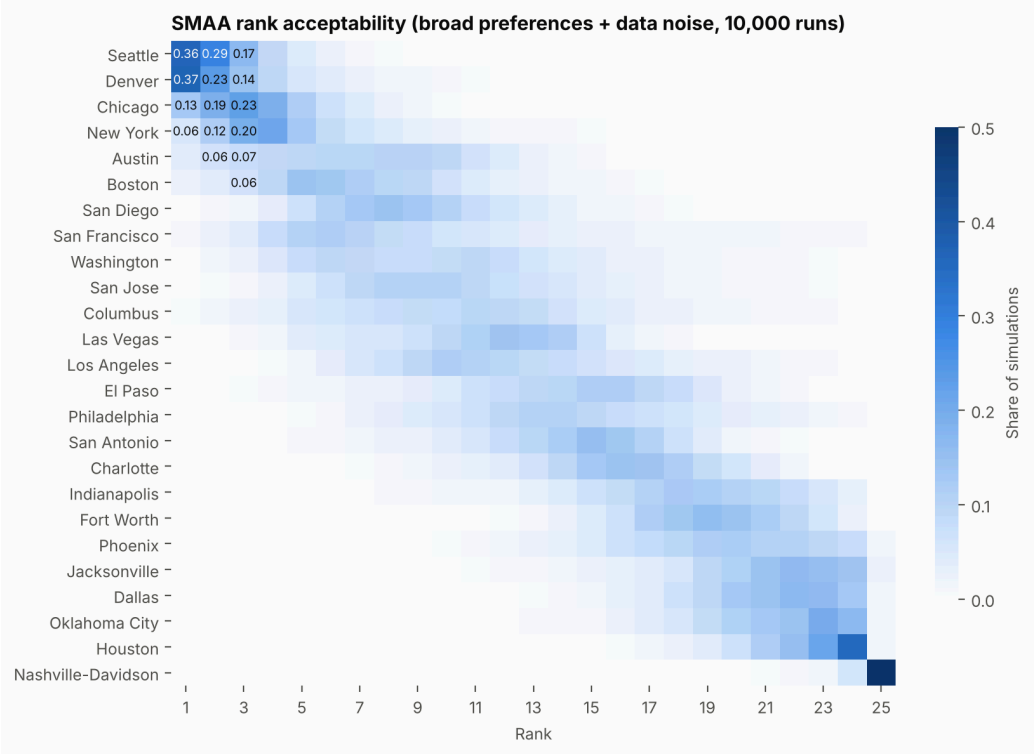


Figure 2. Share of 10,000 SMAA draws (broad preference set plus data noise) in which each city takes each rank.

6.4 Under our stated definition: the final ranking

Adding statements P1–P5 narrows Θ to Θ_S:

- **Fixed pairs:** four pairs now have a necessary order (Chicago and Denver above Nashville; Austin above Dallas and Fort Worth).
- **Possible winners:** six cities can no longer rank first under any preference in Θ_S: Philadelphia, Fort Worth, Phoenix, Dallas, Jacksonville and Nashville.

What the stated preferences weigh. The reference vector θ^0 sits on the edge of Θ_S: it gives education, mobility and affordability the same weight, the least that P1–P4 allow. The average vector of

Θ_S gives education and mobility 0.19 each, affordability 0.11 and the other four domains 0.13 each. Under this single vector Seattle scores 0.812, Denver 0.796, New York 0.785 and Chicago 0.782.

Table 1 gives the final ranking by mean SMAA rank under Θ_S with data noise. With each city's central weight vector fixed and only the data varying, Seattle wins 97.5% of draws (its confidence factor); Denver's is 68.9%.

Table 1. Final ranking (stated preferences Θ_S, 10,000 draws with data noise). "Ref." is the reference scorecard rank; regret is on the 0–1 value scale. San Francisco fails a floor and is listed without a rank; the simulation includes all 25 cities, and leaving San Francisco out changes no other city's position.

Rank	City	Mean rank	First	Top 3	Max regret	Ref.	Floors ($80k)	Status
1	**Seattle**	1.48	63.6%	97.9%	**0.154**	2	pass	**Best city**
2	Denver	2.72	23.5%	75.0%	0.210	1	pass	co-leader
3	New York	3.28	6.0%	60.3%	0.238	4	pass	top-3 contender
4	Chicago	3.64	5.8%	46.7%	0.206	3	pass	co-leader (no-weak-spot rule)
—	San Francisco	6.54	0.4%	6.9%	0.259	8	**fail**	fails rent floor, not ranked
5	Austin	6.91	0.6%	8.0%	0.332	6	pass	
6	Boston	7.09	0.1%	3.0%	0.273	5	pass	
7	San Diego	8.12	0.0%	0.5%	0.295	7	pass	
8	Washington	8.74	0.1%	1.6%	0.319	9	pass	
9	Los Angeles	9.96	0.0%	0.0%	0.320	13	pending (air)	
10	San Jose	10.52	0.0%	0.0%	0.328	10	pass	
11	Las Vegas	11.99	0.0%	0.0%	0.390	12	pass	
12	Columbus	12.85	0.0%	0.0%	0.426	11	pass	
13	San Antonio	14.17	0.0%	0.0%	0.415	16	pass	
14	Philadelphia	15.18	0.0%	0.0%	0.407	14	pass	cannot rank first
15	El Paso	15.71	0.0%	0.0%	0.436	15	pass	
16	Charlotte	16.65	0.0%	0.0%	0.446	17	pass	
17	Indianapolis	18.49	0.0%	0.0%	0.453	18	pass	
18	Fort Worth	19.03	0.0%	0.0%	0.469	19	pass	cannot rank first
19	Phoenix	20.04	0.0%	0.0%	0.447	20	pending (air)	cannot rank first
20	Dallas	20.44	0.0%	0.0%	0.472	21	pass	cannot rank first
21	Houston	21.75	0.0%	0.0%	0.459	24	pass	
22	Jacksonville	21.93	0.0%	0.0%	0.501	23	pass	cannot rank first
23	Oklahoma City	22.82	0.0%	0.0%	0.507	22	pass	
24	Nashville	24.95	0.0%	0.0%	0.608	25	pending (crime)	cannot rank first

Why Seattle. Seattle has no weak domain except affordability. It has the highest advanced-course enrollment (54.2 per 100 students against a median of 32.8) and school sports participation (49.9 per 100 against 25.9). Its price-adjusted teen minimum wage ($19.17 per hour) is the highest of the 25, its students ride free, and it reaches 419,000 jobs by transit in 45 minutes (median 102,000). Its suspension rate is low (2.7 per 100 against 6.9).

Why the others trail.

- **Denver** is held back by violent crime (978 per 100,000) and a lower district graduation rate (71.8%).
- **New York** offers by far the most transit access (1.8 million jobs) but is expensive (rent 41.3% of income).
- **Chicago** is cheap for its quality (rent 25.7% of income) but has fewer school health staff.

Leave-one-statement-out check. Removing any single statement P1–P5 keeps Seattle first in SMAA (54.5–65.5%) and keeps it the minimum-regret city. P1 and P2 both limit affordability; removing them together still leaves Seattle first (55.7% against Denver's 29.2%). The four necessary pairs all come from P5; without it the count returns to 0.

ELECTRE III. With the reference weights, the kernel is Chicago alone, followed by Denver and Seattle. With 400 sampled weight vectors from Θ_S, Chicago is in the kernel 50% of the time, Seattle 41% and Denver 24%. No measure vetoes Seattle against any city. Chicago's strength under this rule is that it has no deep weakness in any measure.

7. Significant Revisions to the Model

We built a complete first model (M0) with the indicators and methods above, then audited every measure against its raw records. Two findings in real data led to revisions.

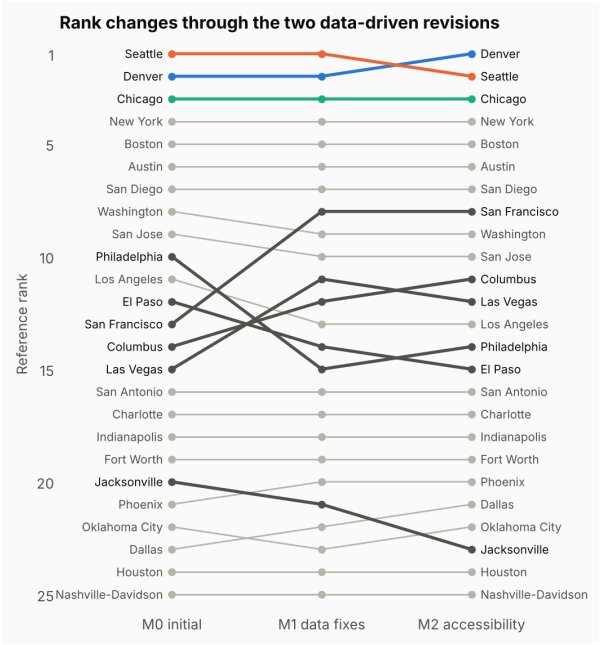


Figure 3. Reference rank of each city through the two revisions. Colored: top three; dark gray: cities that moved three or more places.

Revision 1 (M0 → M1): fixing two measures that mismeasured students' conditions.

- *Evidence 1.* M0 used the EPA's transit-frequency field (trips per hour near home). In San Diego, 96% of residents live in block groups where this field is coded −99999, yet only 22% live more than 0.75 mile from a stop. The code marks missing schedule data, not missing buses. Federal NTD data show San Diego's operator runs 1.11 service hours per resident, above the 25-city median of 0.85.
- *Evidence 2.* M0 averaged graduation rates over all schools in the city. In San Francisco, one charter school for adult re-entry students holds 36% of the city's graduation cohort and graduates 4%, pulling the city to 56%. Oklahoma City (one charter school, 51% of the cohort, 48%) and Las Vegas (21%, 7%) show the same pattern. A resident student enrolls by address in district schools, not in these programs.
- *Change.* We replaced transit frequency with NTD service hours per resident plus weekend service retention. Graduation now uses non-charter schools only, which places San Francisco at 85.4%. The other five school measures keep charter schools, which enroll resident students; Section 8.3 tests applying the same rule to them.
- *Controlled comparison.* Changing graduation alone moves San Francisco 13 → 9. Changing transit alone moves it 13 → 10; together, 13 → 8. Las Vegas rises 15 → 11 and Philadelphia falls 10 → 15. The top three do not change (Kendall τ between M0 and M1 = 0.893).

Revision 2 (M1 → M2): from service supplied to places a student can reach.

- *Evidence.* Service hours count buses, not destinations, and correlate 0.895 (Spearman) with transit access. Scoring both would count transit twice.
- *Change.* We replaced service hours by the EPA's 45-minute transit accessibility. This field counts jobs reachable from each block group by walking and scheduled transit, computed from GTFS

timetables. Jobs stand in for shops, libraries, workplaces and other destinations.

- *Effect.* Kendall τ between M1 and M2 is 0.967. Denver and Seattle swap places in the reference scorecard. The measure also reveals large within-city gaps (Section 9.3).

Why these revisions should convince more people.

- They come from records, not from a preferred winner: both were found by checking raw files. The reference top three (Seattle, Denver, Chicago) was the same in M0, M1 and M2; only the order of the first two changed in M2, by 0.0005.
- Each fix makes a measure describe what a resident student can actually use: the bus schedule, the school she would attend, the destinations she can reach.

8. Why the Ranking Is More Than Our Opinion

8.1 Fixed reference points make the ranking stable when the pool changes

Popular objective methods such as entropy weighting derive weights from how much the measures vary among the candidates. Their results therefore depend on who else is in the pool. We tested this directly: removing one city at a time and re-scoring the other 24 reverses **90** pairs under min-max entropy weights. Under our fixed reference points it reverses **0**, because a city's score depends only on its own data. The final ranking in Table 1, by mean SMAA rank, also reverses 0 pairs when any one city is removed. Entropy weighting would also put the largest weight on transit access (0.134) simply because New York is an outlier, and it would crown New York.

8.2 Different aggregation rules, similar leaders

Rule (Kendall τ with our reference ranking)	Leader
Linear value, no diminishing returns (0.873)	Seattle
Smooth exponential value, k = 1, 2, 4 (0.927, 0.933, 0.867)	Denver
Geometric mean of domains, CES ρ = 0 (0.880)	Denver
Low substitution between domains, CES ρ ≤ −0.85 (0.75 at ρ = −1)	Chicago
Mobility–activities complementarity, 2-additive Choquet (0.987)	Seattle
ELECTRE III with vetoes (kernel)	Chicago
Entropy weights / CRITIC (0.733 / 0.807)	New York / Seattle

The same three cities lead under every rule except entropy weighting. Which of the three comes first depends on how much a strong domain may make up for a weak one. Seattle or Denver lead when trade-offs are allowed; Chicago leads when they are not.

8.3 Sensitivity

- **Weights (Figure 4).** Raising any one domain while scaling the others shows where the leader changes. Seattle leads once education reaches 0.148, inclusion 0.163 or mobility 0.238. Denver leads once affordability exceeds 0.143 or health 0.138. Chicago leads once safety exceeds 0.263. The first four tests start from the reference weights θ^0, where Denver and Seattle are tied, and are then repeated under Θ_S.
- **Nodes.** Shifting every reference point randomly by up to ±10% of its range (2,000 runs) keeps the winner among Denver (64.6%), Seattle (32.0%) and Chicago (3.5%). With weights drawn from Θ_S, Seattle wins 60.0% of shifted runs and Denver 27.0%.
- **Data error.** Multiplying every standard deviation by 0.5, 1, 2 or 4 keeps Denver and Seattle as the two most frequent winners (Seattle 43.1%, 44.5%, 40.8%, 31.8%).

- **One measure at a time.** Removing any one of the 20 measures leaves the champion Denver 13 times, Seattle 6 times and Chicago once. Under Θ_S (each domain keeps its weight), Seattle stays first for 19 of the 20 removals. Removing the graduation rate makes Denver first (65%): Seattle's lead depends on Denver's low district graduation rate (71.8% against 87.2%).
- **Data intervals.** Placing every value at the unfavorable end of its 90% interval removes all four necessary pairs of Θ_S. Those orders depend on point estimates, so we do not claim them as certain.
- **Alternative data choices.** Each of the following keeps Seattle, Denver and Chicago on top:
  - Chicago's 2021 police-department crime definition (960 instead of 516 per 100,000) moves Chicago from 3rd to 4th.
  - Metro-area air monitors put Seattle first.
  - Applying the non-charter rule of Revision 1 to all five school measures (charter schools enroll 5% of Seattle's students, 23% of Denver's and 50% of Oklahoma City's) keeps Seattle first in 57% of Θ_S draws against Denver's 31%. Also leaving out alternative schools gives 43% against 37%.
  - Deflating the teen wage by non-housing prices, so that housing cost counts only through rent, leaves the reference top two unchanged and Seattle first in 63.3% of Θ_S draws.
  - A hierarchical Poisson–Gamma estimate [24] of traffic deaths at ages 14–18 shrinks small-city rates toward the 5.96 per 100,000 average (overdispersion $\chi^2 = 116.4$ on 24 df) and keeps Seattle first ($\tau = 0.947$).

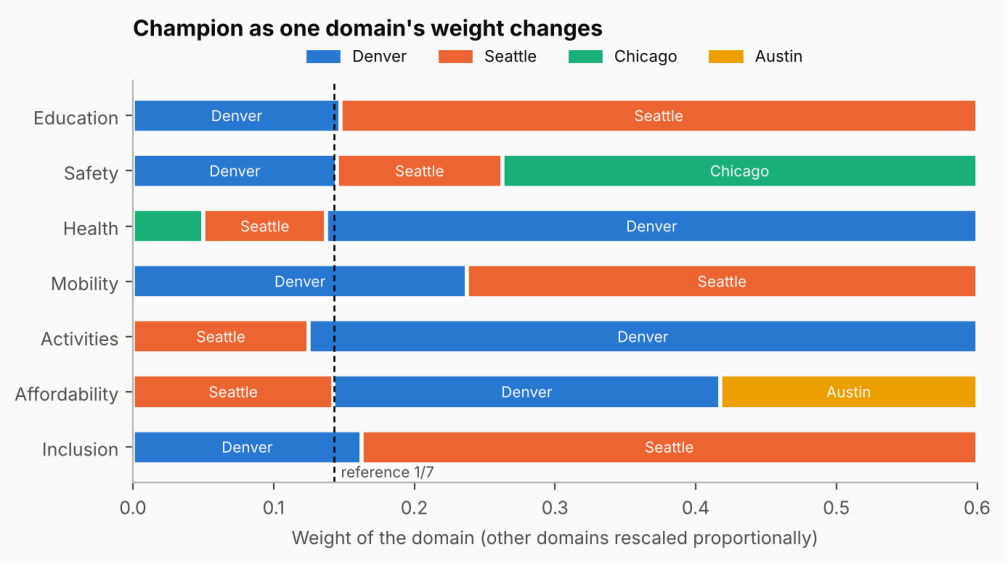


Figure 4. Leader as one domain's weight changes (others rescaled proportionally). The dashed line is the equal weight 1/7.

8.4 External checks

We compared our results with outcomes we did not use to build the model, including the 2023 Youth Risk Behavior Survey (YRBS) [23].

External outcome (not used in the model)	Compared with	Spearman ρ	Expected sign
YRBS 2023: students who skipped school because they felt unsafe (9 cities)	violent crime	+0.73	+ ✓
YRBS 2023: students not on a sports team (7 cities)	school sports participation	−0.93	− ✓
YRBS 2023: students who felt sad or hopeless (11 cities)	our health domain	−0.76	− ✓
YRBS 2023: students bullied at school (11 cities)	our inclusion domain	+0.45	− ×
GTFS 2026 timetables: weekend/weekday trips (23 cities)	NTD weekend retention	0.73	+ ✓
Opportunity Atlas: adult income of children from low-income families [21]	total value U	0.32 (p = 0.12)	+ ✓ (weak)

Three of four youth outcomes agree with our measures in the expected direction. Bullying does not: city non-discrimination policy measures what a city promises, not what happens in school hallways. We report this as a limit of the inclusion domain.

8.5 How much agreement would settle the ranking?

Figure 5 shows how the number of fixed pairs grows as preferences are narrowed. Most of the disagreement lies **inside** domains (how to split, say, safety between crime and traffic), not between domains. If the split inside domains is fixed at the reference, 95 of 300 pairs are already settled. If, in addition, every domain weight is held within ±25% of 1/7, 224 pairs are settled. In the opposite case, with every domain weight fixed at exactly 1/7 and the split inside domains left free, only 5 pairs are settled. Agreeing on how each domain is measured settles more of the ranking than agreeing on how much each domain matters.

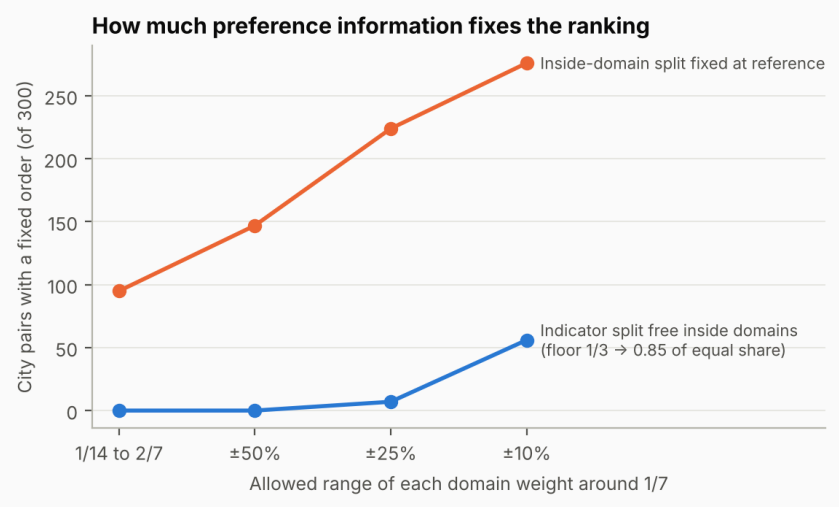


Figure 5. City pairs whose order is fixed for every preference in the set, as the set is narrowed.

9. Who Should Choose Which City

9.1 Student profiles

The model answers different questions for different students by changing both their floors and their weights (doubling the weight of their priority domain).

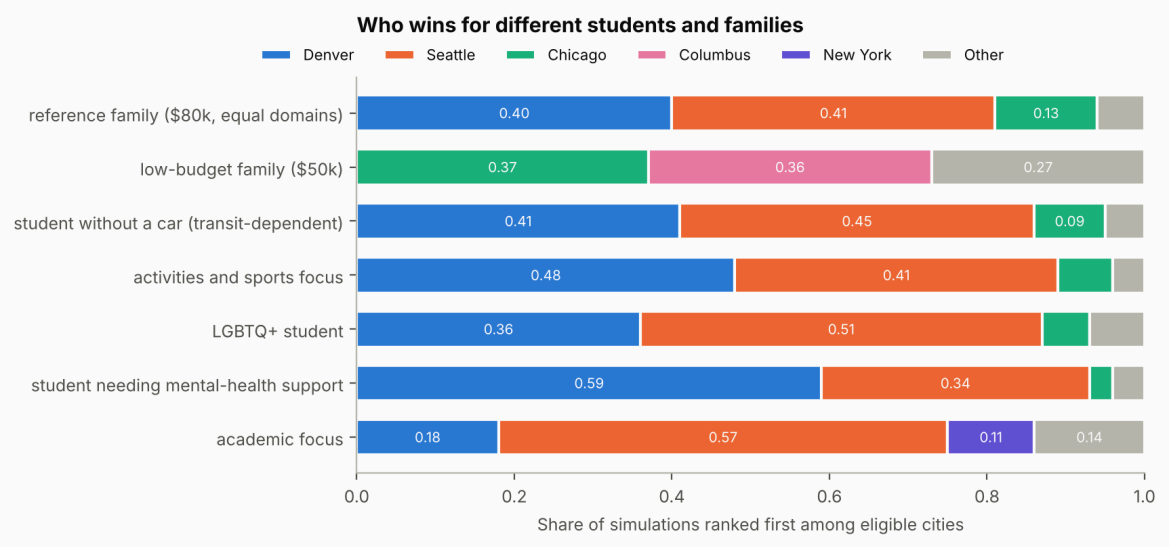


Figure 6. Share of 5,000 draws in which each city ranks first among the cities that meet each profile's floors.

Profile	Extra floor	Cities eligible	Leader
Academic focus	—	24	Seattle (0.57)
LGBTQ+ student	city HRC policy score ≥ 75	19	Seattle (0.51)
Student without a car	≤ 25% of residents without any transit access	18	Seattle (0.45)
Activities and sports	—	24	Denver (0.48)
Needs mental-health support	≥ 5 youth facilities per 100,000 under 18	14	Denver (0.59)
Low-budget family ($50,000)	rent floor at $50,000	17	Chicago (0.37), Columbus (0.36)

9.2 Income changes the eligible set

The rent floor depends on income, so the eligible set does too:

- At **$120,000** all 25 cities pass.
- At **$70,000** San Jose and San Francisco fail.
- At **$50,000** eight cities fail: New York, Los Angeles, San Diego, San Jose, San Francisco, Seattle, Denver and Boston. The best remaining cities at reference weights are Chicago, Columbus, Austin, Las Vegas and Washington.

At reference weights, the leader among eligible cities is Seattle at $100,000 and above, Denver at $70,000–$80,000 (by less than 0.001 at $80,000) and Chicago below $70,000. Under our stated preferences Seattle leads at $80,000 (Table 1).

9.3 Inside the city: averages hide who is left out

City averages assume that a student lives in a typical neighborhood (Assumption 2). We tested this with census-tract data. We mapped ACS 2020–24 tract rents [22] to the transit-accessibility block groups (98–100% population coverage) and asked what share of residents live in tracts that are **both** affordable on $80,000 (rent at most 30% of income) **and** reach at least 60,000 jobs by transit within 45 minutes.

City	Affordable at $80k	Affordable and transit-accessible ($80k)	Same at $50k
Philadelphia	94.6%	94.3%	38.1%
Chicago	87.5%	86.1%	36.5%
Denver	68.2%	68.1%	6.6%
Seattle	38.5%	38.5%	1.1%
San Francisco	15.5%	15.5%	5.9%

- **Seattle's averages describe its central neighborhoods.** Fewer than four in ten residents live where an $80,000 family can afford rent and a student can reach the city by transit.
- **Replacing the access measure with this joint share keeps Denver first** at reference weights (τ = 0.953) and strengthens Chicago's case for middle-income families. Under $\boldsymbol{\Theta_S}$ Seattle stays first, but by less: 47% of draws against Denver's 38%.
- **Some cities strand their least-served residents.** The 20% of residents with the least transit access reach fewer than 500 jobs by transit in Jacksonville, Fort Worth, Oklahoma City, Nashville, Charlotte, Indianapolis, Las Vegas and San Diego.

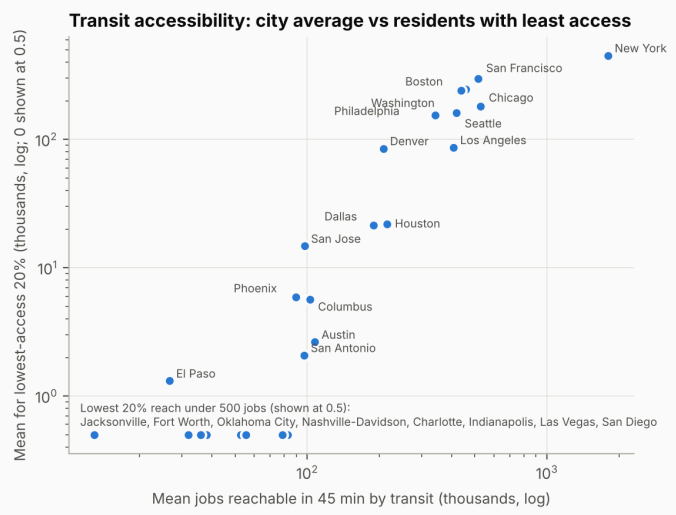


Figure 7. Mean transit access against the mean for the 20% of residents with the least access (log scales).

10. Strengths and Weaknesses

Strengths

- **Floors before trade-offs.** No strength can compensate for unaffordable rent or high risk, and borderline cities are flagged as pending instead of being passed or failed silently.
- **Exact robustness, not just one weighting.** Linear programs test every pair over the whole preference set. SMAA turns that into frequencies with a standard error of at most 0.005, and minimax regret serves cautious families.
- **Stable when the pool changes.** Fixed reference points mean that adding or removing a city reverses no other pair.
- **Data-driven revisions.** Two revisions came from errors found in real records, with controlled comparisons.
- **Reproducible.** Every number in this paper comes from archived public data and our scripts.

Weaknesses and what would fix them

- **Proxy for destinations.** The accessibility measure counts jobs, not schools, parks or teen programs. Point-level routing to actual facilities with current timetables would measure student trips directly.
- **Inclusion is measured by policy.** HRC points describe what a city promises; school-level climate surveys for all 25 cities would measure students' experience.
- **School location is not school access.** We use schools located inside each city; attendance-zone and lottery rules would show which schools a student can actually attend.
- **Some measures are dated.** Graduation rates are for the 2020–21 cohort and accessibility uses 2017 jobs with 2020 schedules. Seattle's lead over Denver depends on graduation: removing it makes Denver first under Θ_S (Section 8.3).
- **Stated preferences are ours.** P1–P5 follow our definition. A family with other priorities can use the profile results or change the statements.

11. Conclusion

1. **Criteria.** We measured 20 conditions in seven separate domains that matter to a high schooler who cannot rely on a car, and left out nine criteria with reasons.
2. **Ranking.** We ranked all 25 of the largest U.S. cities. Under our definition, **Seattle is the Best City for a High Schooler**: first in 63.6% of 10,000 simulations, top three in 97.9%, smallest worst-case regret (0.154). It passes every floor for an $80,000 family.
3. **Close contenders.** Denver is a near-tie on equal weights (0.8210 vs 0.8205) and leads when affordability or health matter more or when graduation is left out. Chicago leads whenever no weakness may be traded away, and is the best value for families earning under $70,000.
4. **Merit beyond opinion.** The leaders survive equal weights, data errors, node shifts, five aggregation rules and the removal of any city or measure. Three of four independent youth outcomes agree with our measures.
5. **Revisions.** Two revisions from real data replaced a broken transit field, removed adult-school distortions from graduation rates, and moved the model from buses supplied to places reachable.

Seattle's lead is real but narrow, and it belongs mostly to families who can afford its connected neighborhoods. The next step is to measure what students can reach from where they can afford to live.

Seattle Tops Our Ranking — But Your Family's Budget Writes the Fine Print

We compared the 25 biggest U.S. cities on what a teenager without a car can actually reach, afford and use. Seattle came out on top, with Denver and Chicago close behind.

Picture a 15-year-old after the last bell. Practice ends at 5:30. Can she take a bus home, or does someone have to leave work to drive her? Can she sign up for AP Chemistry at her neighborhood school? Will a counselor see her if she is struggling? Those questions, not nightlife or house prices, decide whether a city is good for a high schooler.

How we judged. Our team gathered public records for the 25 most populous U.S. cities and turned them into 20 measures in seven areas. *Education* covers advanced courses, graduation and counselors. *Safety* covers violent crime and traffic deaths. *Health* covers clean air, school nurses and psychologists, and youth mental-health clinics. *Getting around* covers how many places a bus can reach in 45 minutes, walkability, student fares and weekend service. *Things to do* covers parks, teen library programs, school sports and the minimum wage for a first job. *Cost* covers rent and prices. *Fair treatment* covers anti-discrimination laws and how often students are suspended.

Some things can't be traded. Before scoring anything, we checked four basics: rent below half of an $80,000 family income, and violent crime, traffic deaths and polluted-air days below danger levels. A city that fails one is not recommended, however good its schools are. San Francisco missed the rent line: a two-bedroom costs about half the family's income.

Everyone weighs things differently. One family cares most about safety; another about cost. So instead of picking one set of weights, we let a computer try 10,000 different ones that fit our definition, and counted how often each city came out first.

The result. Seattle won 64% of the time and finished in the top three 98% of the time. Its students ride transit free, a student can reach about 420,000 jobs by bus and train within 45 minutes, and its high schools have about one school-sport spot and one advanced or college-credit course enrollment for every two students. Denver came second, cheaper and with more school health staff, but more violent crime. New York and Chicago followed.

Top 5
1 Seattle · 2 Denver · 3 New York · 4 Chicago · 5 Austin
San Francisco would be 5th but fails the rent check for an $80,000 family

We changed our minds twice. Our first model used a federal "buses per hour" figure. When we checked the raw files, it was blank for almost all of San Diego, wrongly making the city look busless. We also found that one adult-education school was dragging San Francisco's graduation rate down to 56%; the schools local teens actually attend graduate 85%. Fixing both moved San Francisco from 13th to 8th. Then we stopped counting buses and started counting places a student can reach. These fixes came from the data, not from wanting a different winner, and the top three never changed.

Why trust it? We tried the ranking every way we could think of: equal weights, bigger data errors, different scoring rules, dropping any one city or measure. Seattle, Denver and Chicago stayed on top every time. In the cities with student surveys, places we scored as safer had fewer students skipping school out of fear, and places with more school sports had more students on teams.

The fine print. Seattle's strengths sit in neighborhoods many families cannot afford. Only about 4 in 10 Seattle residents live where an $80,000 family can afford rent *and* buses reach plenty of destinations, compared with 86% in Chicago. If your family earns closer to $50,000, Chicago and Columbus are the better bets.

The bottom line. The best city for a high schooler is one where she can reach good classes, safe streets and things to do on her own, and where her family can afford to live. Today that is Seattle for families who can afford it, and Chicago for many who cannot.

How we built the ranking, in four steps

1. Measure
20 things a teen can actually use, from public records for 25 cities

→

2. Check the basics
rent, crime, traffic deaths and dirty-air days must stay below danger lines

→

3. Score
early gains count most: going from few to some courses beats going from many to more

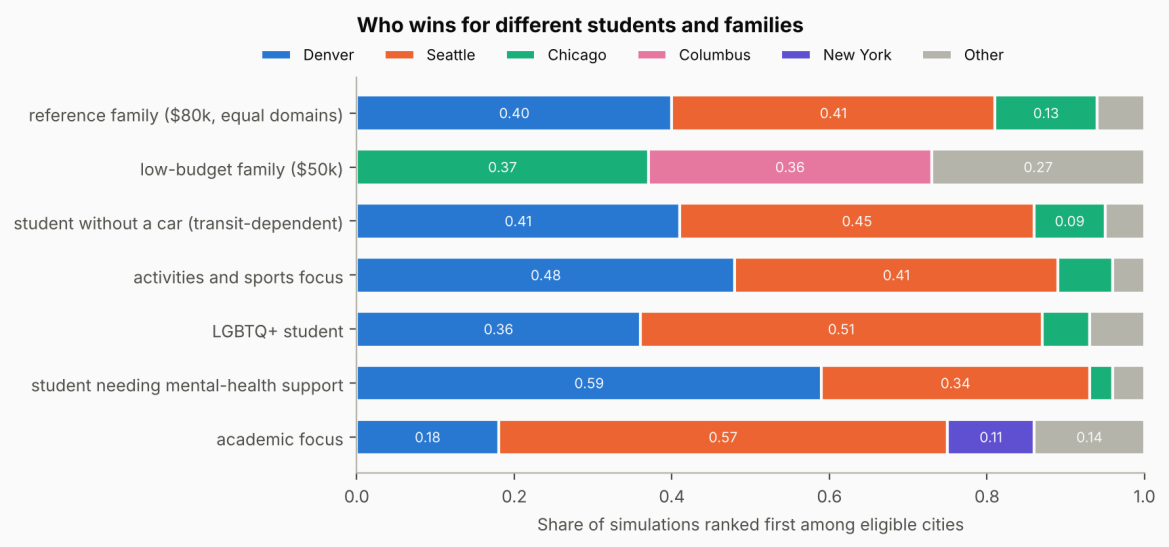
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4. Stress-test
10,000 different weightings plus data errors; count who wins

Rank	City	How often #1	Top 3
1	Seattle	64%	98%
2	Denver	23%	75%
3	New York	6%	60%
4	Chicago	6%	47%
5	Austin	<1%	8%
6	Boston	<1%	3%
7	San Diego	<1%	<1%
8	Washington, D.C.	<1%	2%
9	Los Angeles	0%	0%
10	San Jose	0%	0%

What the numbers mean. "How often #1" is the share of our 10,000 test runs in which a city came out on top. A city that wins under many different weightings is a safer bet than one that wins only if you value one thing above all else.

Pick by your priority. Families who rank cost first should look at Chicago and Columbus. Students who need mental-health support or love activities will find Denver on top. Students focused on academics, or who want strong anti-discrimination protection, will find Seattle first.



Who wins for different students: share of test runs in which each city ranks first, among the cities that meet each student's basic needs.

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Report on Use of AI Tools

This report is not counted toward the 25-page limit.

Our team used AI assistants throughout the project. All final choices of definition, criteria and the five preference statements P1–P5 were reviewed and confirmed by the team; every number in the paper was produced by code that the team can rerun from archived public data.

#	Tool	How we used it	What we did with the output
1	OpenAI ChatGPT / Codex (local agent with sub-agents), Nov. 2026	Searching for and downloading public datasets for 25 cities (Census, CRDC, EDFacts, FBI, FARS, EPA, SAMHSA, NTD, HUD, BEA, IMLS, HRC, fare and wage pages); writing collection, cleaning and archiving scripts; recording access failures	Checked file hashes, row counts and coverage tables; kept failed downloads as documented gaps instead of filling them
2	Anthropic Claude, used through Claude Code (cloud sessions), Nov. 2026	Reading 92 Outstanding-paper analysis cards and the HiMCM writing guides; proposing model structures; writing the model code (indicator matrix, floors, piecewise-linear value model, linear programs, SMAA, regret, ELECTRE III, sensitivity, figures); auditing data; drafting this paper and the news article	Ran all code ourselves; verified LP bounds against 3,000 sampled weight vectors, hand-checked one city's score, compared rerun results with stored results; reviewed and edited the text and chose the final definition and preference statements

Representative prompts (paraphrased).

1. "Collect the most recent public data for the 25 largest U.S. cities on high-school course access, school support staff, crime, traffic deaths, air quality, youth mental-health services, transit, parks, libraries, wages, rent and non-discrimination policy; record sources and gaps." → datasets, scripts and a gap log (tool 1).
2. "Read the knowledge-base paper cards and propose a model that ranks cities without letting low rent compensate for safety, and that shows how robust the ranking is to different weights." → the floors + piecewise value + LP + SMAA design (tool 2).
3. "Build the indicator matrix from the archived data and run all models; audit any measure that looks wrong." → code and the two real-data revisions in Section 7 (tool 2).
4. "Write the full paper and a one-to-two-page news article from the results files, following the HiMCM page limit and the writing guides." → the draft of this document; every number in it was checked against the results files (tool 2).

Limits we observed. The AI tools sometimes proposed weaker choices that we corrected after checking: an initial ELECTRE veto threshold that vetoed almost every comparison, a sampling loop that ran out of memory, and a sentence that misdescribed course enrollments as a share of seats. None of these errors remains in the results.